

Marco Vetrano



About me

Computational Physicist and Data Scientist. Specialized in developing scalable **Machine Learning** models for extracting insights from data. I design and implement **AI** solutions to solve complex analytical problems.

Areas of Expertise

Physics • Machine Learning
Network Analysis • Data Analysis

marco.vetrano01@gmail.com

Marco Vetrano

Marco Vetrano01

EXPERIENCE

04/2023-03/2026	Research and Development in Quantum Machine Learning UNIVERSITY OF PALERMO · Italy 📍 Development of Classical-Quantum algorithms for data compression and extraction of physico-chemical properties from large astrophysical datasets. The codes were primarily developed in Python using NumPy, PyTorch, and Scikit-Learn for model implementation and pre-processing, and Matplotlib and Seaborn for visualization. Furthermore, the codes were deployed on Leonardo's proprietary HPC and IBM Quantum devices. Skills: Python Programming, Machine Learning, Linux, Git	
02/2025-08/2025	Research and Development in Quantum Machine Learning IFISC - UNIVERSITY OF THE BALEARIC ISLANDS · Spain 📍 Implementation of classical and quantum machine learning models for chaotic time series prediction. Deployment of codes on IFISC's proprietary HPC. Skills: Python Programming, Machine Learning, Linux, Git, Docker	

EDUCATION

2023-2026	Ph.D. in Physics · University of Palermo 🏛️ Ph.D. research carried out at the University of Palermo, with a 6-month period abroad at IFISC. Scholarship funded by PNRR Funds on the STILES project.
05/2026 - Present	Master in AI Development and Engineering PROFESSIONAI · 🏛️ Dual master's degree offered by ProfessionAI with final attainment of a certification with European validity. The master lasts two years and aims to train in the development of AI solutions using custom models, with mixed Python and C++ programming; it provides instruction in Computer Vision, Explainable AI, and Reinforcement Learning.
2020-2023	Master's Degree in Physics · University of Palermo 🏛️ Master's Degree in Physics specializing in Theoretical Information Physics and Quantum Machine Learning.
2016-2019	Bachelor's Degree in Physical Sciences · University of Palermo 🏛️

LANGUAGES

Italian	C2	Native Speaker	
English	C1	Writing	Reading
		C1	C1
		Listening	Speaking
		C1	C1

CERTIFICATIONS

2026	Udemy	Certification: Machine Learning and Data Science in Python
2023	Duolingo	C1 Certification in English

TECH STACK

Python	SQL	C++	Docker	Java
Hugging Face	REST API	XGBoost		
Agent Orchestration	Linux	AWS		
Prompt Engineering	Quarto	LaTeX		
	Git	PySpark	PyTorch	

PUBLICATIONS

2025	"State estimation with quantum extreme learning machines beyond the scrambling time.", Springer Nature npj Quantum Information 11.1 (2025): 20.
2025	"Exoplanetary atmospheres retrieval via a quantum extreme learning machine.", arXiv preprint arXiv:2509.03617.

PROJECTS

2025	QuantumToolBox UNIVERSITY OF PALERMO · Italy 📍 A Python package developed primarily using the NumPy, Scikit-Learn, and Plotly libraries for the implementation of quantum dynamics and the implementation of classification and time series prediction models. The package is available on GitHub in the QuantumStuff repository	
2025-2026	AstroQELM UNIVERSITY OF PALERMO · Italy 📍 A Python package designed for end-to-end analysis of spectral data. The package features functions for reading and parsing h5 and txt data, pre-processing and compression with Scikit-Learn, and processing and classification with Qiskit, NumPy, and Scikit-Learn. The package is available in the AstroQELM repository	
2026	Instagram-Analyzer · Italy 📍 Backend app developed with FastAPI in Python for scraping Instagram profiles. The app uses an Instagram username entered by the user, from which it extracts the last uploaded reel and its associated comments using two Apify actors, and calculates its BPM and duration. It also includes two Hugging Face AI models for Speech-to-Text video transcription and sentiment analysis of the comments. The app is fully containerized and available on GitHub.	
2026	Olist-App · Italy 📍 Project for the end-to-end analysis of the Olist e-commerce dataset, freely downloadable from Kaggle. The package, fully containerized with Docker, contains functions for downloading data, building the database with SQLAlchemy, querying, aggregating, and cleaning data in PostgreSQL. It also contains three sentiment analysis models of increasing complexity, developed with Scikit-Learn, PyTorch, and Hugging Face. Currently under development, it will in the future contain a graphical interface developed with Streamlit, a customer care agent, and a Tableau interface for visualizing aggregated data with SQL.	
2026	CopyWriter AI Agent · Italy 📍 An AI agent created with LangGraph and LangChain. The agent is set up to write SEO articles, perform keyword research via SerpAPI, and use them to write 600-word economic-financial articles, generating a title and sections. The article is then used to generate metadata and converted into HTML format for publishing as a draft on WordPress via REST API.	

TALKS

2023	"Information Scrambling in Quantum Extreme Learning Machines for State Estimation" at the New Trends in Non-equilibrium Statistical Mechanics 2nd Course
2024	"QELM response to scrambling dynamics and applications to atmospheric retrieval" at the International School on Non-Equilibrium Phenomena - Quantum Reservoir Computing Working Party
2025	"Exoplanetary Retrieval with Quantum Extreme Learning Machines" at the National Congress of the Italian Physical Society

AWARDS

2023	First place at the QuHack4AI: Quantum Hackathon for Industrial Applications , organized by the University of Bologna and Unipol with a project on variational algorithms for the study of hydrogen and water molecules
2023	"Eduardo Gugino" Award for the Master's thesis on "Information Scrambling in Quantum Extreme Learning Machines" for the extraordinary session of March 2023.

OTHER EXPERIENCE

- | | |
|---------------|---|
| 2022 and 2024 | Academic Tutor (STEM)
UNIVERSITY OF PALERMO · Italy 📍
Teaching support for Physics, Mathematical Analysis, and Geometry courses, with personalized mentoring including for students with learning disabilities. |
| 2022 | Data Analyst Intern
RI.MED FOUNDATION · Italy 📍
Application of Network Analysis techniques for the study of RNA structures, identifying critical interaction patterns. Skills: Network Analysis, Data Visualization, Data Cleaning and Analysis, Python Programming |
| 2018 | Data Analyst Intern
IASF - INSTITUTE OF SPACE ASTROPHYSICS AND COSMIC PHYSICS · Italy 📍
Development of C scripts for the processing and spectral analysis of data from black holes. Skills: C Programming, Data Analysis and Visualization. |

Marco Vetrano ✉ marco.vetrano01@gmail.com 📍 Palermo, Italy ☎ +39 3459460312
@ https://marcovetrano01.github.io/Portfolio/

I authorize the processing of my personal data present in the CV in accordance with Legislative Decree 30 June 2003, no. 196 and the GDPR (EU Regulation 2016/679)